

CSE 420 – Compiler Design
Quiz - 02
SET B

Student Name: _____

Student ID: _____

Section: _____

Total Marks: 15

Consider the following grammar, which represents a simplified C while loop. **Assume, Stmt is the start of the grammar symbol.**

$$\begin{aligned} Stmt &\rightarrow \textbf{while} (Cond) ; \\ Cond &\rightarrow \textbf{id} \mid \varepsilon \end{aligned}$$

1. Construct the LR(0) Automaton of the grammar. (5 Marks)
2. Construct the SLR(1) Parsing Table. (5 Marks)
3. Simulate the SLR(1) parsing algorithm for the following input string.

while() ;

Show the stack, remaining input and parser action at every step. (5 Marks)